

Orbit Modules

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How to tailor Orbit

Why Orbit

Orbit needs tailoring. That means it takes a little bit of effort. It also means that there's no excess cloth. Unlike other generic systems like GURPS (which I adore, to be clear, no shade), the tailoring aims to still be founded on minimalism. This should result in a fast building experience for the GM, and a simple, but expressive system for the players.

Flexible

The core (or the [star](#) , if you will) of Orbit is the resolution system. It links with the other systems through the metacurrency, which you are free to name for your setting. Different subsystems use different names for it and you're free to use them or take them as inspiration.

Hackable

It's very easy to design a subsystem, or hack away at one to get something more to your liking, and in fact I'd encourage it!

How to tailor Orbit

First, take the resolution system.

Now, you can think about what you need: do you need magic? Guns? Horse-riding? Latch onto the system the magic subsystem, the guns subsystem, the horse-riding subsystem. Not there? Invent one. The simplest subsystem might just be a skill. Let me explain.

Take for example horse-riding. We can **choose** to make it a very complicated subsystem, with three custom skills, one for caring, one for taming wild horses, and one for riding. Or we can choose to keep it simple, as I did in the example Western system, and just have riding be a **Ride** ability.

There's a few systems you probably **need** to add:

- a damage / hp system (player characters **will** get hurt eventually, even in a game with no combat)
- a basic skill system
- an initiative system (doesn't need to be linear, can be reaction based like the one in western)

Why not add them by default? I hear you ask?

Well, a damage / hp system can be more or less lethal. It can rely on hp or wounds. A skill system can be granular or paint in broad strokes. I'll provide modules, but it's up to you whether your game needs the one or the other.

Mechanically, how do I tailor Orbit

The system is written in typst. You can pretty much just concatenate the contents of the files (ie: copy paste the texts) and it all should pretty much work, to be honest. I'll work to try to make it as seamless as possible. Contributions are welcome. Or you can just take the pdfs of different submodules and use those separately.

Resolution System

Introduction

The system uses a 2-die roll-over mechanic based on Abilities. Each character has 6 abilities (see below).

Characters also have Push Points (the game's metacurrency. The actual name depends on the rest of the system this gets latched on to), which allows them to push themselves beyond their normal limits.

Main roll

The base roll is simple:

- Roll two dice of the indicated size
- Compare them with the DC:
 - If both succeed, it's a full success: you get what you want and nothing bad happens
 - If both fail, it's a full failure: get ready for consequences (most commonly in combat, the enemy gets to do something)
 - If one succeeds, it's a mixed success: you get what you want, but you suffer some consequence (meaning that both the success and failure effects take place)
- If the two dice have **the same value**, then it's a **strong** result: whichever effect occurs, it's going to be amplified

Risk roll

Sometimes a PC might want to not care about contingencies and go straight for the metaphorical kill, even at their own risk. This is the **risk roll**. Roll 1 die (subject to advantage), and treat it as if it were a pair of dice for the purposes of the main roll's degrees of success: any result is automatically a **strong** result, and there's no mixed success.

Ability dice

Ability scores are tracked by increasing die size. The starting point for all abilities is d6.

Assigning Abilities

Based on character creation choices, base ability scores can change. For example, an excellent marksman might start with a d8 or d10 in Aim instead of the usual d6. Conversely, a simple farmhand might start with a d4 in Charisma.

d4 ← d6 → d8 → d10 → d12

Push Points and Pushes

Each character has a pool of Push Points. A Push Point can be used to **push oneself**, letting the character behave, **for a moment** (a single roll, a single action), as if they were better at the skill. This means, most notably, increasing the die size by one step following the usual progression:

d4 → d6 → d8 → d10 → d12

If a die is going to be pushed beyond d12, choose one:

- spend one point and gain a Push die
- spend 2 points at once to replace the d12 with a d20. Further pushes beyond d20 create a Push die.

Push dice are rolled alongside the main roll, and their result is added to the result of both dice in the main roll. Each die may be separately subject to advantage. A Push die starts at size d4. Extra Push Points can be used to increase its size, using the progression outlined above.

You can double the effect of a push¹ (with GM consent) by introducing a complication. Instead of a complication, the World can tally a due.

The World (personified in the GM) can keep a tally of the PC's dues. A sort of symbol of the unfair luck they might have had, or of the favours fate handed them. Fate, however, doesn't do gifts, and luck turns around.

Dues

Dues—accumulated within Bargains with Fate—are tokens the GM gets to spend to insert world or NPC moves while outside of the timing defined by the normal resolution system. For example, the GM might spend a token to:

- add a complication to a scene
- have an NPC interrupt a player turn / sequence of player turns

Generally, Difficulty Classes are:

Difficulty	DC	When to Use
Easy	2+	Simple tasks
Pretty Hard	4+	Routine challenges for competent folk
Oomphy	6+	Significant obstacles
Incredibly Hard	8+	Expert-level tasks
Legendary	10+	Heroic feats

¹making N paid points behave as $2N$. (e.g. $1 \rightarrow 2$, $2 \rightarrow 4$)

Favorable circumstances may grant **advantages** or **disadvantages**.

Advantages and Disadvantages

Advantages and disadvantages can stack, or cancel each other out². The final number of kept rolls is the same number of dice as the original unmodified roll. Advantage / Disadvantage applies to any additional dice (like push dice) at the GM's discretion.

Advantage: Add a die of identical size to the one being rolled, after any pushes. Take the highest results. **You can convert advantage into one push up to (and including) d8, two pushes up to d12, or three pushes after reaching d20.**

Disadvantage: Add a die of identical size to the one being rolled, after any pushes. Take the lowest results.

As with Gumshoe³, characters are presumed to be **competent**. No rolls are needed for activities that any denizen of the Game's World would be able to do. Additionally, as ability levels increase, this minimum baseline for success also rises.

Example Core Skill list

Any RPG generally will need skills to handle social encounters and basic interactions with the world.

The only skills that I believe are **needed** to cover the bases are **Strength** , **Dexterity** and **Perception** . They are the core of what a character needs to tell the resolution system in order for it to work.

Beyond those, I believe strong arguments can be made for **Charisma** , **Intelligence** , **Luck** , and **Endurance** . I personally use them, however it can be argued that Charisma and Intelligence should be left to the players, Endurance can be folded into strength, and Luck can be left to the dice. It's your choice!

Naturally, all names can be changed to better fit the vibe of the setting. For example, I renamed Charisma to "Gift of the Gab" in the Western Example Setting.

²more advantages or disadvantages stack together, advantages and disadvantages cancel each other out

³one of the touchstones of this system, with Ironsworn, YZE and DH

Initiative System

Ways to do Initiative

Initiative can be based on fixed order or reactions.

Fixed Order

At the beginning of time-sensitive situations, call for a roll for initiative. The roll can use any appropriate stat within your system. The stat you prefer for stealth to determine the routs after an initial ambush, for example. You can either make NPCs symmetrical (you'll need statblocks, however), or let the players roll their preferred stat against the NPC's DCs.

For when there's no particular conditions, use an existing proxy for reaction speed within your system, or add one if you don't have it.

You then play in the fixed order, forming rounds akin to those in DnD and Pf2e.

Characters can choose to delay their action to any point later than their initial roll. I recommend letting it happen at no penalty (eg shifting their position in initiative, like pf2e does), as it's just an extra tool for creativity that doesn't break the narrative.

Reaction System

The PCs start (unless they're surprised). If they partially succeed or fail on a roll, the GM gets a reaction, which either means a turn for an NPC, or an environmental effect. If all rolls are player-facing, PCs will likely roll to counteract. This can lead to reaction chains, where a PC keeps reacting in a very dynamic fashion. To prevent this from becoming a spiral, other characters are allowed to break the reaction chain as soon as the first "round" of reactions is over.

Example

Bob has a d8 in Aim and tries to hit NPC Alice. Alice is a bandit wanted in 20 towns, so she has a legendary DC of 11. Knowing this, Bob spends two Sand Points to push his roll to a d12. Unfortunately, he rolls a 6 and a 10, thus taking a full failure. Alice now gets to shoot back at him. Bob rolls Awareness to try to find cover to mitigate the damage. He acts quickly, having no real alternatives, calling for a [risk roll](#). He rolls his d6 against a simple DC of 5 because they're in a saloon full of barrels: it's a 6, so he manages to dive, taking no damage. He emerges unscathed from the situation, but is now cornered by Alice. How will he proceed?

Bob could instead choose to react with Grit. He heroically takes the bullet in the shoulder, [risk](#) rolling his d10 in Grit against a legendary DC of 11. He uses a Sand

Point and manages to succeed by rolling an 11. Eventually he'll have to stop (he does have lead in his shoulder), but for now he doesn't give ground.

It's reasonable for the GM to impose disadvantage on reaction rolls using the same ability that just failed (barring special circumstances).

In a reaction, you can perform quick actions that grant advantage if they present a complication.

Example 2

Alice is fleeing from the police when she sees a man recognize her from across the street. She approaches him to convince him not to call [the law](#), but her Gift of the Gab isn't working (she rolled a 3 and a 2). The man now runs toward the sheriff. Alice reacts using her Gift of the Gab again, but the GM rules that she's rolling with disadvantage. Alice could cancel her disadvantage by drawing her gun and pointing it at the man to frighten him. Naturally, she's in town and there's people around. It might just not be worth it.

After the first exchange occurs, the player might be tempted to keep the chain of events going: as their reaction triggers a reaction from the NPC, and so forth. Other players [can](#) and probably should intervene (unless it's fun for them to watch as their friend keeps digging their own grave with more and more complications. In that case, have fun!).

Wound System

Why use wounds

Wounds make for very dynamic damage systems, with damage having instantaneous effect on the characters. This can give better immersion, where hp create some dissonance because a character is as effective at 1 hp as they are at 200.

Wound System

There's many ways to get hurt in a given setting. The more you get hurt, the more you are wounded. This is represented with different wounded **levels** . In the example Western System, there's 5 levels of wounded, from **unharmed** to **dead** .

Grit

The metacurrency, (called **Grit** in this subsystem) allows you to fight your wounds, or to recover.

- you can spend 1 Grit to reduce the damage you take by 1 level.
- you can spend 1 Grit to heal 1 level of wound.

The GM might decide that one might need to roll an ability for the effect to apply in particularly gritty settings (as in the case of the example Western system). This should be clearly communicated to the players.

Wound levels

There are 5 wound levels by default: **Unharmed** , **Wounded** , **Badly Wounded** , **Dying** , **Dead** .

For each level, the PC incurs a mechanical penalty. In this document, I'll reduce their ability die sizes. How many die sizes to detract per wound level is the GM's choice, but I'll provide a couple of examples below.

Example 1

The 5 levels are:

Condition	Mechanical Effect	Description
Unharmed	No penalties	You're in fighting shape, ready for whatever comes next.
Wounded	All abilities reduced by 1 die size	You're hurt but functional. That bullet graze or knife cut is slowing you down, but you can still act.
Badly Wounded	All abilities reduced by 2 die sizes. Some abilities may become unusable (GM discretion).	You're in serious trouble. Blood loss, broken bones, or severe trauma make even basic actions difficult.
Dying	All abilities reduced by 3 die sizes. You fall unconscious and cannot act unless spending Grit to briefly regain consciousness for one action by pushing .	You're at death's door. Your vision fades, strength leaves you, and staying conscious requires tremendous effort.
Dead	Character dies, but may make one final action with any ability at -4 die size before expiring (eg dump all Grit into one final shot).	Your time has come. In your last moments, you might manage one desperate act of will before the darkness takes you.

When taking damage, you can move from one level to another. Strong results, like strong failures in getting out of the way of a bullet, might also move the PC by two levels. (eg unharmed → badly wounded)

In this variant, healing requires expending two Grit.

Note to GMs: this system gets [very](#) lethal [very quickly](#). Pay careful attention to telegraph potentially game-ending encounters.

Example 2

Example 1 is pulled 1 for 1 from Gumslingers, and that system was very lethal. This alternative is a bit more heroic.

Condition	Mechanical Effect	Description
Unharmed	No penalties	You're in fighting shape, ready for whatever comes next.
Wounded	One ability of you or the GM's choice is reduced by 1 die size	You're hurt but functional. That bullet graze or knife cut is slowing you down, but you can still act.
Badly Wounded	One ability of you or the GM's choice is reduced by 1 die size. Some abilities may become unusable (GM discretion).	You're in serious trouble. Blood loss, broken bones, or severe trauma make even basic actions difficult.
Dying	One ability of you or the GM's choice is reduced by 1 die size. You fall unconscious and cannot act unless spending Grit regain consciousness by pushing , or being woken up.	You're at death's door. Your vision fades, strength leaves you, and staying conscious requires tremendous effort.
Dead	Character dies, but may make one final action with any ability before expiring (eg dump all Grit into one final shot).	Your time has come. In your last moments, you might manage one desperate act of will before the darkness takes you.

When taking damage, you can move from one level to another. Strong results, like strong failures in getting out of the way of a bullet, might also move the PC by two levels. (eg unharmed → badly wounded)

The same ability **can** be reduced more than once. Ideally, player and GM will agree on which ability should be reduced, with the player proposing and ability and the GM accepting it. The GM, however, can overrule the player's proposal.

Resting Subsystems

Exhaustion

The party is already going to want to rest because they will want to heal and regain points. However, for campaigns needing it, you can use exhaustion:

Exhaustion

Exhaustion comes in levels, each carrying a mechanical penalty. When the highest level is reached, the character is simply fully unconscious and will sleep for a full day.

In a nutshell, each level of exhaustion carries a mechanical penalty. A good starting point is that each level carries a 1d penalty to an ability, and lowers the maximum Push Point pool by 1.

Recuperating Push Points

Push points can be recuperated. How? That depends on the genre and setting, naturally. You can tie it to consumables (i.e. potions, food), and I recommend tying fixed, safe regeneration to resting in safe conditions. Here's a baseline:

Getting a full night's rest restores points, depending on the conditions:

- in a safe environment with proper supplies → full set of points
- without supplies or unsafe → Half points

In the latter case, you may choose to take a level of exhaustion to recover your whole set of points.

You can also recover points by:

- resting briefly while consuming a meal (1 point)
- using consumables (1 point⁴)
- resting for an extended amount of time (About 1 point per hour in-game)

As with all things, you can tweak things to your liking.

In the case of consumables, you might want to tie them to a roll. For example, you might tie drinking liquor, or smoking to gaining points, and have a character roll a stat (like intelligence, for their analytical mind to punch through the foginess, or Endurance to avoid running out of breath).

An example on the mechanical effects might be:

Result	Effect
--------	--------

⁴stronger / rarer consumable can provide more, though I wouldn't recommend making it too common

Strong failure	The stat receives a –2d penalty
Full failure	The stat receives a –1d penalty
Partial success	Recover 1 point, the stat receives a –1d penalty
Full success	Recover 1 point
Strong success	Recover 2 points

Alternatively, you could simply take out the stat penalty component, make the result fully binary between nothing happening mechanically and the 1 point recovery, and use the degree of success for narrative elements inside the scene. That much is your prerogative as the GM. You can also do both, though that might feel very punishing.

Healing wounds

Much like for push points, healing obviously depends on genre. In worlds with magic, you might have healing magic, and in worlds with medicine, you may have medicine. This is just a structure you can reskin freely, obviously.

Resting

Getting a full night's rest restores wound levels in the following amounts. Roll Endurance (or whichever best proxy for “grit” or “constitution” you have in the system) against a CD of 4.

Result	Effect
Strong failure	You get worse (gain one level)
Full failure	Recover 0 levels
Partial success	Recover 1 level
Full success	Recover 2 levels
Strong success	Recover 3 levels

This set assumes your wound system has 5 levels, going from “unharmed” to “dead”. If you have more levels, keep it proportional. If you have less, provide additional benefits for higher degrees of success.

You can also heal from wounds by having them be treated, or spending push points, as already established in the wounds module.

Treatment

Roll an appropriate stat, like “Intelligence”, for example.

Result	Effect
Strong failure	You get worse (gain 1 level)
Full failure	Recover 0 levels

Partial success	Recover 1 level, but your next roll suffers the mechanical penalties of the old level
Full success	Recover 1 levels
Strong success	Recover 2 levels

Consumables

In a world with magic or very advanced technology, you might have magic potions or wondrous medicine. In such worlds, you may have consumables that heal one⁵ level of wounds.

⁵or more, though that'd be an incredibly powerful effect

Magic System

Objectives

This magic system aims to give players the freedom to handle magic as they wish by being flexible, but sufficiently limiting so that it doesn't get out of hand.

The main tool to achieve this is the metacurrency: **Mana**. As in Vagabond⁶, Mana can be used to modify or enhance spells.

The second element is **skills**. Magic relies on stats, as in many other systems.

Depending on the setting, you may want to change which skills are tied to magic.

Investing metacurrency into magic-tied skills will, as with any other skill, make you better at the kind of magic tied to it, which we can represent by raising the baseline for automatic success, and giving access to more spells of that category.

Mana

Beyond the standard uses for Mana in rolling (it's still the system's metacurrency), there's a few uses for Mana in the system:

- Changing the shape of a spell
- Increasing the range of a spell
- Increasing the area of effect of a spell
- Combining spells

Base Spells

- **Slime:** Coat a target in sticky slime.⁷
- **Slide:** Coat a target in slippery stuff.
- **Acid:** Coat a target in corrosive acid.
- **Bolt:** Throw a bolt of any Element or Energy at a target, dealing damage.
- **Manipulate:** Manipulate any Element or Energy.
- **Telekinesis:** Move things as if they weren't at a distance.⁸
- **Telepathy:** Talk to other beings at a distance, silently. They can answer.
- **Mind-reading:** Read minds. Might be spotted.

This is a non-exhaustive list. If your setting needs more spells, feel free to add them.

Elements and Energies

There's 4 default elements and 4 default energies.

Elements

- **Fire**
- **Water**
- **Air**
- **Earth**

Energies

- **Electricity**
- **Heat**
- **Sound**
- **Light**

⁶which is a major touchstone, alongside SWADE, for this magic system

⁷Targets for coating must be at most of around your size, e.g. a creature of your size, or a small area of around your own surface area.

⁸using your magic skill instead of your otherwise more appropriate skill for moving stuff

Possible groupings

These spells should be tied to one or more skills, depending on the level of granularity of the magic system in the setting. If we have one skill, the grouping is trivial. In this document, we'll have 4 groups: **Coating** , **Elemental** , **Energetic** , **Mental** .

Elemental and Energetic might look small, but note that **Bolt** and **Manipulate** are [very](#) flexible spells.

Coating	Elemental	Energetic	Mental
Slime	Manipulate Element	Manipulate Energy	Telekinesis
Slide	Bolt of Element	Bolt of Energy	Telepathy
Acid			Mind Reading

Combining Spells

You can use **Mana** to combine the effect of spells you know. These **Composite Spells** will share in the effects of their sources. For example, you might combine **Slime** + **Manipulate Fire** to get a persistent, adhesive burn. Or you might combine **Telepathy** with **Manipulate Light** or **Manipulate Sound** to temporarily blind or deafen a guard without actually emitting light or sound. Or you might combine **Slime** or **Slide** with **Telepathy** to make a thought particularly slippery, or sticky in one's mind.

Character Creation

Objectives

The character creation system pretty much boils down to fleshing out the assumptions the resolution system is built upon to allow for variety in power levels, while preserving build variety.

Character Creation

The core concept is that you invest Push Points to obtain permanent die size increases.

Character Creation

The cost to permanently increase an ability is:

Die Step	Cost	Descriptors
d6 → d4	−1 point*	Not great
d6 → d8	1 point	Pretty great
d8 → d10	2 points	Incredible
d10 → d12	3 points	Peak human ability

*You can lower an ability to d4 to gain one Push Point.

The numbers are adjustable, but I find this works pretty well already.

Budgeting

Depending on the desired power level, you'd set different starting budgets. A good rule of thumb for a starting point is 2 points per ability score present. With six abilities, that's 12.

Assuming 6 abilities, good defaults are:

Tone	Budget
Gritty	9
General	12
Heroic	15

In the western example, I used 8, because I wanted a lethal, quick and gritty setting and vibe. Play around with some builds yourself before telling the players your budget to see what they can build.

Templates

It's a good idea to provide templates. Once you've set a budget, make some example builds.

The examples below use six names from the example core skill list in the resolution system: Strength, Dexterity, Perception, Charisma, Intelligence, and Endurance. In a finished system, replace the ability names and the metacurrency name with the ones used by that setting. If the system uses Luck as a seventh ability, add it to the ability list and adjust the budget. The number shown for Push Points is the remaining rechargeable limit after buying the listed ability changes from the d6 baseline.

Gritty Budget: 9

Hardcase Scout

Spend 5, keep 4

Push Points

4

Abilities

Strength

d8

Dexterity

d10

Perception

d8

Charisma

d4

Intelligence

d6

Endurance

d8

Notes

Excellent under pressure, but socially brittle and short on reserves.

Silver Tongue Survivor

Spend 4, keep 5

Push Points

5

Abilities

Strength

d4

Dexterity

d6

Perception

d8

Charisma

d10

Intelligence

d8

Endurance

d6

Notes

Talks first, reads the room well, and avoids direct physical contests.

General Budget: 12

Field Medic Spend 8, keep 4		Duelist Spend 7, keep 5	
Push Points	4	Push Points	5
Abilities		Abilities	
Strength	d6	Strength	d8
Dexterity	d6	Dexterity	d10
Perception	d10	Perception	d8
Charisma	d8	Charisma	d10
Intelligence	d10	Intelligence	d4
Endurance	d8	Endurance	d6
Notes		Notes	
A focused support specialist who solves problems with expertise and steady hands.		Strong in tense confrontations, less useful when patience or study matters.	

Heroic Budget: 15

Paragon

Spend 10, keep 5

Push Points

5

Abilities

Strength

d10

Dexterity

d10

Perception

d8

Charisma

d8

Intelligence

d8

Endurance

d8

Notes

Broadly exceptional, with enough reserves to push through important scenes.

Mastermind

Spend 12, keep 3

Push Points

3

Abilities

Strength

d4

Dexterity

d6

Perception

d12

Charisma

d10

Intelligence

d10

Endurance

d8

Notes

Dominates planning and social leverage, but has little room for brute force or repeated pushes.

Character Name:

Gender:

Budget:

Archetype:

Push Points

Abilities

Strength

Dexterity

Perception

Charisma

Intelligence

Endurance

Conditions

☐ Unharmed

☐ Impaired

☐ Wounded

☐ Dying

☐ Out

Notes

Equipment